

tific. Many traditional songs indicate that certain people are masters of knowing or understanding the language of crows. It is a belief that all living things other than humans possess a sixth sense and therefore interpret sounds made by different animals and birds. However, without any scientific basis, these beliefs have been termed as superstition.

While living with animals and birds, humans may recognise the intentions, emotions or thoughts behind the sounds made by these animals and birds, even if the sounds themselves are not totally understood. These interpretations can be based on eye gaze, facial expression, vocalisation, body posture (including movement of bodies and limbs) and gustatory communication (scent, pheromone and taste).

At the primitive level, humans communicate with animals using vocalisation, hand signals, body postures and touch to express love and care. Artificial intelligence (AI) can help humans express feelings in a more effective way. This may soon become a reality as scientists are making use of electronics through the implementation of AI to decode the different dialects of animals.

There have been continuous attempts to create translation devices that can interpret animal sounds and their meanings. While some such attempts have failed or managed to simply mimic animal sounds, others have seen some success, translating a few sounds. Although, these endeavours have been exciting and novel, these have not reached the ability to truly translate what is going on inside the mind of animals. Much of this technology is being used to study animals in the wild, but someday soon it may be parlayed into a device for humans to talk to the animals around them.

Progress made towards animal communication

The man credited with leading the charge in decoding animal sounds, Dr Constantine Slobodchikoff, is an expert in animal referential communication studies using prairie dogs as a model species. Dr Slobodchikoff

has been able to index the sounds made by prairie dogs to synthesise their language into English using electronics with AI to maintain and develop the catalogue and to further allow the computer to learn on its own to translate the animal talking to a great extent.

Dr Slobodchikoff does not hesitate to mention that a bias exists among biologists and linguists. He believes that animals are only able to communicate and cannot express emotions, or have conversations based solely on instinct. As per consensus, there is no universal agreement on a common level of consciousness across all animal species, but scientists who have studied the gorilla named Koko have proven that apes are more than capable of thought and feeling.

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Over the years, researchers have recorded hundreds of hours of prairie dog calls using hidden microphones. A sophisticated combination of electronics and AI is used to analyse each recording by looking at how different frequencies and overtones stack on top of one another. This is how researchers have learned that the calls can be clustered into different groups, with each cluster having its own signature set of frequencies and tones.

Dolphins have always been one of the most fascinating animals for scientists to study, as they are considered to be the second most intelligent animal on Earth, next to humans. The

first effective animal translator was Cetacean Hearing and Telemetry (CHAT) translator, which was used to communicate with dolphins. Using this translator, a team of scientists studying a particular group of dolphins decoded whistles and their meanings, and CHAT translated the dolphins' whistles to the word.

Further, a company is developing a program using AI analysis software along with CHAT to decipher what dolphins speak. The software for decoding human language is currently used to collect information about the type of emotion a speaker is exuding, and the program has already mastered 40 human languages. Companies interested in this utility believe that decoding dolphins will further this capability.

The same technology is also being used to translate the sounds of other animals. Sounds made by animals have different patterns and intonations tied to different concepts and words. Scientists feed these to a computer that maps out their meaning and context. So far, the program has recorded with refinement, distinguishing hundreds of different gibbon calls.

Although having an AI translator does not necessarily mean humans will be able to have a heart-to-heart conversation with their pets, as there are vast differences between human and animal cognition, and humans are a long way from understanding the latter. However, this kind of communication technology could help humans better understand animals and their behaviour—this would mean more than just forging closer emotional ties with them.

Simple communication with animals could eliminate guesswork in caring for animals and even save their lives. Similarly, AI-based electronics communication technology could make things easier for farmers and ranchers, for instance, by quickly identifying animals that are sick by detecting signs of pain on their faces.

Role of AI

Growing success of neural networks in applications like speech recogni-